LESSON 01 / 06

One known solution generates a second

Normalize a linear equation as $y'' + P(x)y' + Q(x)y = 0$ and suppose y_1 is a known nonzero solution on an interval. Put $y = vy_1$. After using $y_1'' + Py_1' + Qy_1 = 0$,

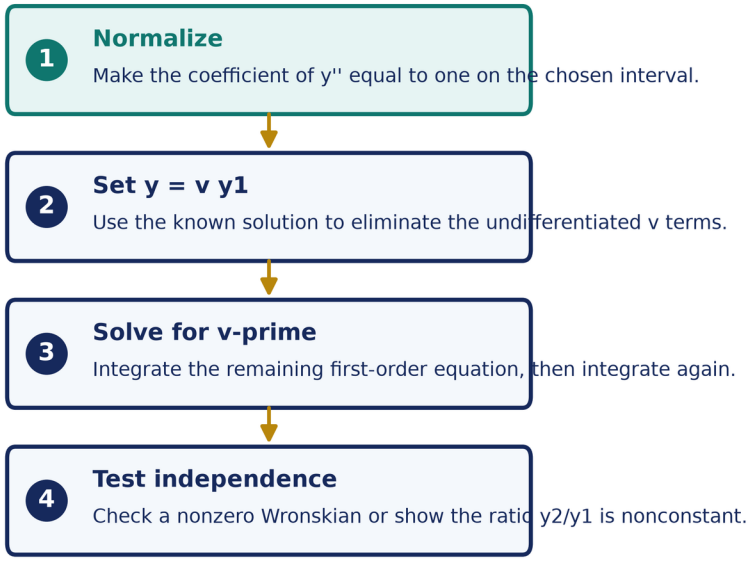
$$y_1v'' + (2y_1' + Py_1)v' = 0.$$

This is first order in $w = v'$. Integrating once more gives the standard formula

$$y_2 = y_1 \int \frac{e^{-\int P(x)dx}}{y_1^2}\,dx.$$

The integral is computed locally where $y_1 \neq 0$; a resulting formula can sometimes be extended by direct verification.

Reduction of order converts a second-order equation into a first-order one



READ THE FIGURE

The Wronskian convention used here is W=y1 y2-prime minus y1-prime y2.

WORKED EXAMPLE

S17 and Q34: a corrected Wronskian sign

For $y'' + y = 0$, $P = 0$ and $y_1 = \sin x$. On an interval where $\sin x \neq 0$,

$$y_2 = \sin x \int \csc^2 x\,dx = \sin x(-\cot x) = -\cos x.$$

Direct substitution extends $-\cos x$ to a solution on $(-\infty, \infty)$. The general solution is $C_1 \sin x + C_2 \cos x$. With $y_2 = -\cos x$,

$$W = y_1y_2' - y_1'y_2 = \sin^2 x + \cos^2 x = +1.$$

The source's -1 is a sign error under this convention; choosing $y_2 = \cos x$ instead would reverse the Wronskian but not the independence conclusion.

WORKED EXAMPLE

P13 and Q35: reduce the Euler equation

Given $y_1 = x$ for $x^2y'' - 2xy' + 2y = 0$, put $y = xv$. Then

$$y' = v + xv', \qquad y'' = 2v' + xv''.$$

Substitution leaves $x^3v'' = 0$, so on $x > 0$ or $x < 0$, $v'' = 0$ and $v = Ax + B$. Hence

$$y = Ax^2 + Bx.$$

For $y_1 = x$, $y_2 = x^2$, the Wronskian is $W = x(2x) - 1(x^2) = x^2$, nonzero away from the singular point $x = 0$.

COMMON TRAP

A Wronskian sign depends on ordering and representative choice

Independence requires a nonzero Wronskian, not a particular sign. Always state the ordering and the actual second function before reporting W.

LESSON 02 / 06

Differentiation splits into straight lines and an envelope

A Clairaut equation has the form

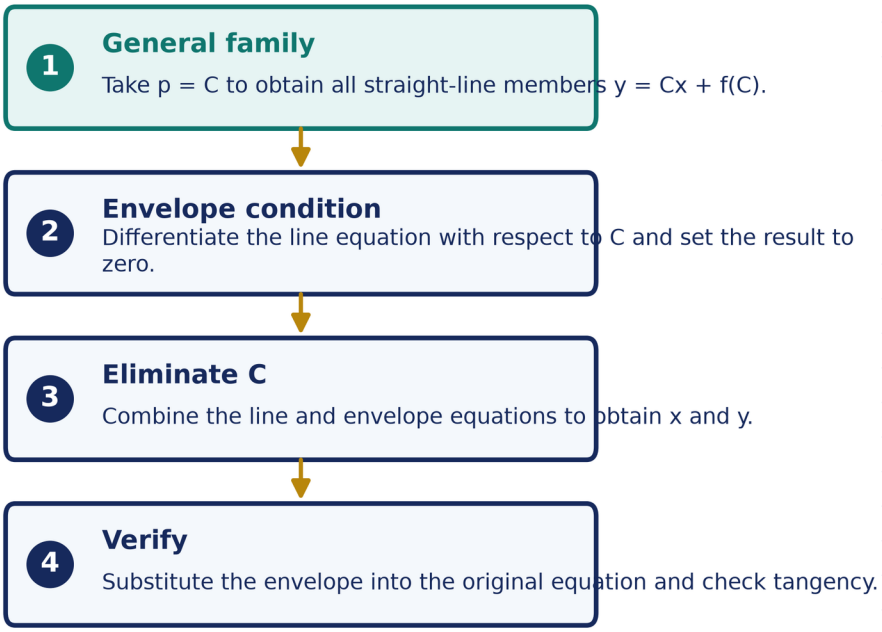
$$y = xy' + f(y').$$

Set $p = y'$ and differentiate:

$$p = p + (x + f'(p))\frac{dp}{dx} \quad \Longrightarrow \quad (x + f'(p))p' = 0.$$

The branch $p' = 0$ gives $p = C$ and the general straight-line family $y = Cx + f(C)$. The other branch, $x + f'(p) = 0$, gives a parametric singular solution; equivalently, eliminate C using $F = y - Cx - f(C) = 0$ and $F_C = 0$.

Clairaut's two branches answer two different questions



READ THE FIGURE
A singular solution need not belong to the general family for any finite value of its arbitrary constant.

WORKED EXAMPLE

SIS and Q32: a downward envelope

For $y = xp + p^2$, the general family is

$$y = Cx + C^2.$$

The other branch is $x + 2p = 0$, so $p = -x/2$ and

$$y = x\left(-\tfrac{x}{2}\right) + \left(-\tfrac{x}{2}\right)^2 = -\tfrac{x^2}{4}.$$

Thus the singular solution is $y = -x^2/4$. For a fixed C, its tangent line touches this parabola at $(-2C, -C^2)$, as shown in Visual Lab 6.

WORKED EXAMPLE

PII and Q33: reversing the sign reverses the envelope

For $y = xp - p^2$, the line family is $y = Cx - C^2$. The envelope conditions are

$$y = Cx - C^2, \qquad x - 2C = 0.$$

Therefore $C = x/2$ and

$$y = \frac{x^2}{2} - \frac{x^2}{4} = \frac{x^2}{4}.$$

Direct verification gives $p = x/2$ and $xp - p^2 = x^2/4 = y$.

Why the extra solution is not contradictory

Differentiating the original equation is reversible only with care: the factorized equation permits both $p' = 0$ and $x + f'(p) = 0$. Reporting only the first branch discards the second. The word singular describes the envelope's relation to the general family, not a blow-up or discontinuity.

COMMON TRAP

Envelope algebra still needs a solution check

The envelope conditions are necessary for the standard smooth family construction. Substitution into the original differential equation confirms that the resulting curve is genuinely a solution.

LESSON 03 / 06

Continuity and Lipschitz continuity have different jobs

For $y' = f(x, y)$, $y(x_0) = y_0$, local continuity of f gives a local existence theorem. Picard–Lindelöf adds a Lipschitz condition in y near the initial point and then guarantees local uniqueness. A continuous partial derivative f_y on a nearby closed rectangle is a common sufficient way to obtain that Lipschitz condition.

Failure of a sufficient uniqueness hypothesis does not itself prove non-uniqueness. The examples below prove non-uniqueness by constructing distinct solutions explicitly.

What the standard local statements do and do not assert

Hypothesis	Conclusion	Caution
f continuous	Local existence	No uniqueness asserted
f continuous + y-Lipschitz	Local uniqueness	Theorem is local
f_y unbounded	Lipschitz test fails	Not a non-uniqueness proof
Two solutions verified	Non-uniqueness	Check joining derivatives

READ THE FIGURE
The examples use real-valued differential equations and check both one-sided derivatives where a formula changes.

WORKED EXAMPLE

S18 and Q36: sqrt(y) permits arbitrary waiting times

One solution of $y' = \sqrt{y}$, $y(0) = 0$ is $y = 0$ for all x. Another is zero for $x \le 0$ and $x^2/4$ for $x > 0$. For the latter,

$$y' = \frac{x}{2} = \sqrt{\frac{x^2}{4}} = \sqrt{y}$$

when $x > 0$, and both one-sided derivatives at 0 are 0. More generally, for any $a \ge 0$, take zero for $x \le a$ and $(x - a)^2/4$ for $x > a$. All satisfy the same IVP.

Since $f_y = 1/(2\sqrt{y})$ is unbounded as $y \to 0^+$, the required Lipschitz condition near zero is absent.

WORKED EXAMPLE

P14 and Q37: a cube-root equation with departure at x = 1

Define $y = 0$ for $x \le 1$ and

$$y = \left(\frac{2(x-1)}{3}\right)^{3/2} \qquad x > 1.$$

On the departing branch, put $u = 2(x-1)/3$. Then

$$y = u^{3/2}, \qquad y' = u^{1/2}, \qquad y^{1/3} = u^{1/2},$$

so $y' = y^{1/3}$. At $x = 1$, both one-sided derivatives are 0, and the initial condition $y(0) = 0$ holds. Here $f_y = y^{-2/3}/3$ is unbounded near 0, again defeating the Lipschitz hypothesis.

COMMON TRAP

Real powers and joining points require domain care

For Q36, sqrt(y) requires y nonnegative. For Q37, the displayed nonnegative departing branch uses a real cube root. In both examples the derivative at the waiting-time endpoint must be checked, not assumed from the formulas on either side.

LESSON 04 / 06

Differentiate with x regarded as a function of p

A Lagrange equation has the form $y = xf(p) + g(p)$, where $p = y'$. On a branch that can be parametrized by p, differentiation with respect to p gives

$$p\frac{dx}{dp} = f(p)\frac{dx}{dp} + (xf'(p) + g'(p)).$$

Away from $p = f(p)$, this is linear in $x(p)$. Constant-p solutions excluded by that division must be recovered separately.

WORKED EXAMPLE

S19 and Q38: the corrected differentiation

For

$$y = xp^2 + 1,$$

differentiate with respect to p:

$$p\frac{dx}{dp} = p^2\frac{dx}{dp} + 2xp.$$

Thus, away from $p = 0$ for the derivation,

$$(1-p)\frac{dx}{dp} = 2x \quad\Longrightarrow\quad \frac{dx}{x} = \frac{2\,dp}{1-p}.$$

Integration gives

$$x = \frac{C}{(1-p)^2}, \qquad y = 1 + \frac{Cp^2}{(1-p)^2},$$

with $C \neq 0$ and p restricted to an interval not crossing 1. Direct parametric differentiation confirms $dy/dx = p$ and the original relation; the displayed family is also valid at an isolated parameter value $p = 0$.

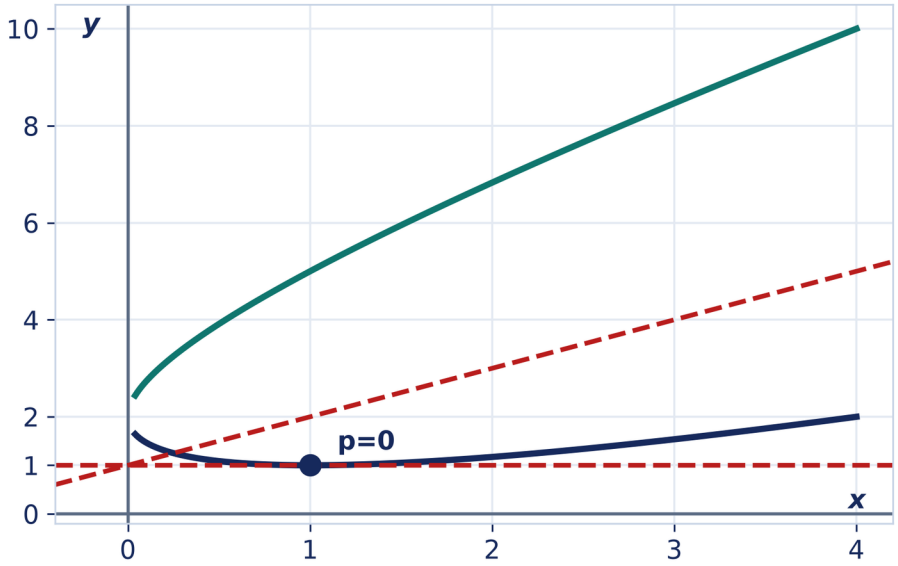
Restore the constant-p solutions

If $p = k$ is constant, then $y = kx + b$. Substitution into $y = xp^2 + 1$ forces $k = k^2$ and $b = 1$. Hence the two additional solutions are

$$y = 1, \qquad y = x + 1.$$

Neither is obtained from the displayed parametric family by a finite nonzero C. These restore the branches omitted by the source's erroneous differentiation and incomplete final list.

Q38 with C=1: two parametric branches and two constant-p lines



READ THE FIGURE

For C=1, eliminating p gives y=x+2 plus/minus 2 sqrt(x) on x>0. The red dashed lines y=1 and y=x+1 are separate solutions; the lower navy branch touches y=1 at (1,1).

COMMON TRAP

The source equation p = 2xp + xp^2 p-prime is not differentiation of y = xp^2 + 1

The product rule first gives p = p^2 + 2xp(dp/dx). Equivalently, the p-parametric calculation above avoids that error and keeps the linear equation in x(p) transparent.

LESSON 05 / 06

One particular solution linearizes the remaining family

For

$$y' = q_0(x) + q_1(x)y + q_2(x)y^2,$$

suppose y_1 is a known particular solution. On a branch with $y \neq y_1$, put $y = y_1 + 1/u$. Subtracting the equation satisfied by y_1 yields

$$u' + (q_1 + 2q_2y_1)u = -q_2.$$

This linear equation supplies the non-y_1 branches. The known solution itself must still be listed separately unless a finite constant happens to reproduce it.

WORKED EXAMPLE

S20 and Q39: y = 0 is not recovered by finite C

Here $y' = y^2 - y$ and $y_1 = 0$. Put $y = 1/u$:

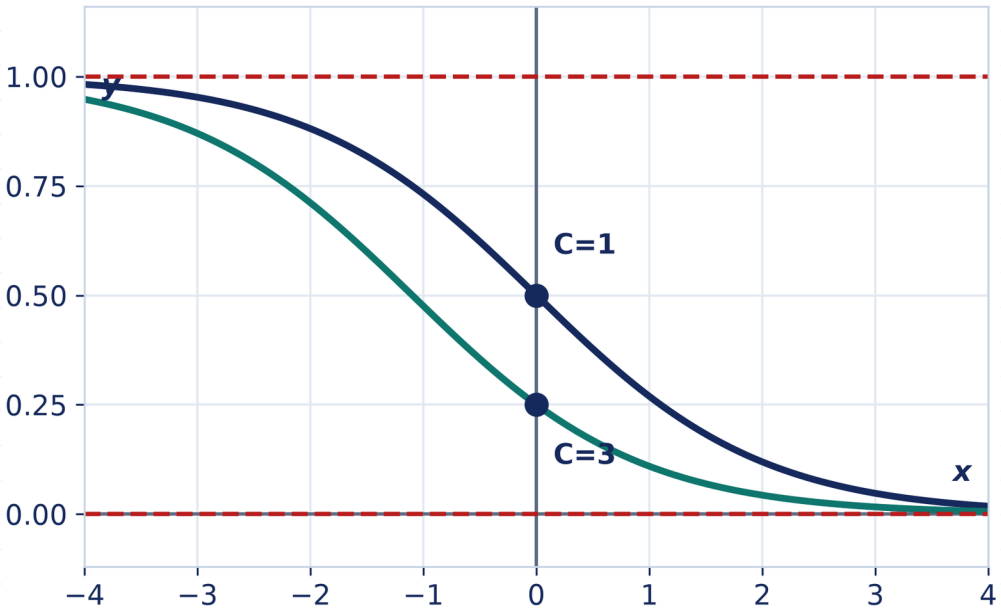
$$-\frac{u'}{u^2} = \frac{1}{u^2} - \frac{1}{u} \quad \Longrightarrow \quad u' - u = -1.$$

With integrating factor e^{-x}, $u = 1 + Ce^x$, so

$$y = \frac{1}{1 + Ce^x}.$$

For $C < 0$, the denominator vanishes at $x = \ln(-1/C)$, so each solution interval must stay on one side of that pole. For $C \geq 0$, the formula is defined on $(-\infty, \infty)$. The separate equilibrium $y = 0$ is not attained for any finite C; $C = 0$ instead gives the other equilibrium $y = 1$.

Q39: positive-C branches connect the two equilibria



READ THE FIGURE

For C>0, 1/(1+C exp(x)) decreases from 1 toward 0. The limiting equilibria are genuine separate solutions; approaching one as C grows is not the same as attaining it for finite C.

WORKED EXAMPLE

P15 and Q40: a Gaussian integrating factor and another omitted equilibrium

For $y' = x(y^2 - 1)$ with $y_1 = 1$, put $y = 1 + 1/u$. Then

$$u' + 2xu = -x, \qquad (ue^{x^2})' = -xe^{x^2}.$$

Therefore

$$u = Ce^{-x^2} - \tfrac{1}{2}, \qquad y = 1 + \frac{1}{Ce^{-x^2} - 1/2}.$$

The separate solution $y = 1$ is not obtained for finite C. The other equilibrium is $y = -1$, obtained with $C = 0$. Every branch must avoid zeros of $Ce^{-x^2} - 1/2$; when $C > 0$ and $2C \geq 1$, these occur at $x = \pm\sqrt{\ln(2C)}$, with the two signs coinciding when $C = 1/2$.

COMMON TRAP

Taking C to infinity is a limit, not a member of the finite-C family

The source uses a large-C limit to motivate the omitted Riccati solutions. The corrected general answer lists those equilibria explicitly and states intervals away from denominator zeros.

LESSON 06 / 06

Add one particular solution to the homogeneous general solution

For $L[y] = ay'' + by' + cy = g(x)$, linearity gives

$$y = y_h + y_p,$$

where $L[y_h] = 0$ and $L[y_p] = g$. Two arbitrary constants belong to y_h. Undetermined coefficients is efficient for exponential, polynomial and trigonometric forcing; variation of parameters is more general.

Choose a particular-solution method from the forcing term

Forcing	Trial or start	Adjustment
exp(kx), nonresonant	A exp(kx)	Find A by substitution
Already homogeneous	Multiply trial by x	Repeat if needed
Polynomial/sine/cosine	Full same-type trial	Keep all related terms
General continuous g	Variation of parameters	Use independent y1, y2

READ THE FIGURE
This extension is additional enrichment beyond the 41 canonical paper questions; its formulas are verified independently in the mathematical suite.

WORKED EXAMPLE

Undetermined coefficients: nonresonant and resonant forcing

For $y'' - y = e^{2x}$, try $y_p = Ae^{2x}$. Then $(4-1)A = 1$, so

$$y = C_1e^{x} + C_2e^{-x} + \tfrac{1}{3}e^{2x}.$$

For $y'' - y = e^{x}$, the plain trial Ae^{x} is homogeneous and gives zero. Instead,

$$L[xe^{x}] = (x+2)e^{x} - xe^{x} = 2e^{x},$$

so $y_p = xe^{x}/2$ and

$$y = C_1e^{x} + C_2e^{-x} + \tfrac{1}{2}xe^{x}.$$

Variation of parameters with the same Wronskian convention

For normalized $y'' + Py' + Qy = g$, let y_1, y_2 be independent homogeneous solutions and $W = y_1y_2' - y_1'y_2$. Then

$$u_1' = -\frac{y_2 g}{W}, \qquad u_2' = \frac{y_1 g}{W}, \qquad y_p = u_1 y_1 + u_2 y_2.$$

The formula follows by imposing $u_1' y_1 + u_2' y_2 = 0$ and matching the first-derivative equation.

WORKED EXAMPLE

A trigonometric interval where cosine is positive

For $y'' + y = 1/\cos x$ on $(-\pi/2, \pi/2)$, choose $y_1 = \cos x$, $y_2 = \sin x$, so $W = 1$. Then

$$u_1' = -\tan x, \qquad u_2' = 1.$$

Hence $u_1 = \ln(\cos x)$, $u_2 = x$, and a particular solution is

$$y_p = \cos x \ln(\cos x) + x \sin x.$$

The interval matters: both $1/\cos x$ and $\ln(\cos x)$ require $\cos x > 0$.

COMMON TRAP

Do not force a resonant trial to work by changing only its amplitude

If the trial solves the homogeneous equation, every amplitude still produces zero on the left. Multiplication by x changes the functional form and supplies the missing forcing term.